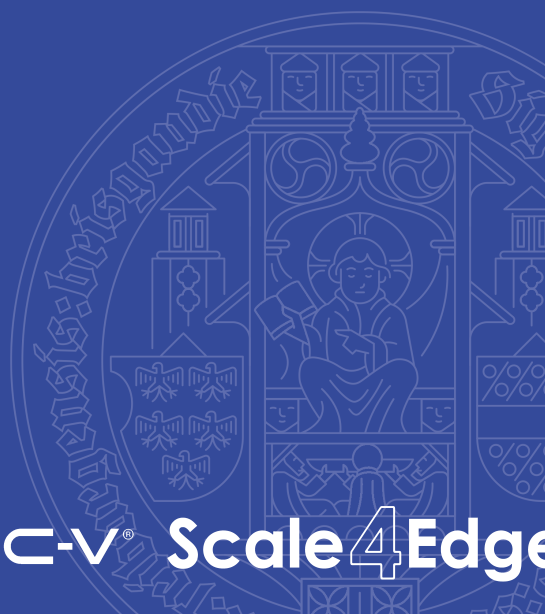


Test and Reliability

Exercise 7: Software-Based Self-Test

Tobias Faller, M.Sc.
Prof. Dr. Ralf Wimmer
Dr. Tobias Seufert

Summer Term 2026



 **RISC-V[®] Scale4Edge**

Exercise SBST:

- 20 lines of code to write
- Expected coverage is 54.21 %
- Improvements:
 - Test for opposite stuck-at faults
 - Test config and status registers
 - Test different patterns for shift registers
 - 75 % coverage achievable

	Fault Coverage	Fault Efficiency	Runtime	Transactions
Full-Scan Stuck-At ATPG	91.86 %	100.00 %	0.01 min	---
Sequential Stuck-At ATPG	88.31 %	96.61 %	20.32 min	---
Functional Stuck-At ATPG	73.31 %	81.78 %	2.64 min	309
Verilator (Simulated)	73.05 %	---	---	309
Full-Scan Transition-Delay ATPG	86.40 %	100.0 %	0.02 min	---
Sequential Transition-Delay ATPG	82.56 %	96.16 %	24.38 min	---
Functional Transition-Delay ATPG	65.58 %	80.66 %	1.24 min	442
Verilator (Simulated)	75.65 %	---	---	442
Verilator (Simulated SA+TDF)	75.46 %	---	---	751